

Final Project Report

GeoAgent: Overcoming Step Degradation in Geospatial Vision-Language Models via Tool-Augmented Agentic Reasoning

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GitHub Repository: <https://github.com/Ali-Abdullah-1921/DL-project>

Mohammad Ali Abdullah (26100241) Muhammad Sheis Hussain (26100220)
Manal Haider (27100024)

Abstract

This paper investigates the problem of step degradation and ontology mismatch in Vision-Language Models (VLMs) during multi-hop spatial reasoning over remote sensing imagery. While domain-specific VLMs perform well on single-step perception, their reasoning accuracy collapses when tasks require sequential logic. To address this, we present an iterative architectural evolution. We propose the “V2 GeoAgent” architecture: an autonomous ReAct loop built upon Qwen2-VL that integrates Oracle-backed vision tools and introduces Recurrent Visual Attention Masking (RVAM). By utilizing targeted Gaussian blurs to physically enforce step-wise image grounding without destroying spatial context, the V2 GeoAgent completely eliminates ungrounded linguistic guessing. Evaluated on a mathematically balanced 600-question spatial benchmark, GeoAgent achieved a massive +31.5% absolute improvement over the baseline on 4-hop reasoning tasks, demonstrating robust, autonomous multi-step logic over complex satellite data.

1. Related Work

The development of GeoAgent builds upon three primary intersections of deep learning: large vision-language models for remote sensing, tool-augmented programming, and visual attention mechanisms.

1.1. Vision-Language Models in Remote Sensing

Recent advancements have adapted generic VLMs (like LLaVA [3]) to the remote sensing domain [1]. However, standard models rigidly compress images into low-resolution squares, causing sub-pixel objects to disappear. The introduction of Qwen2-VL [2] resolved this by implementing Naive Dynamic

Resolution and Multimodal Rotary Position Embeddings (M-ROPE). Despite these upgrades, evaluating VLMs reveals they still suffer from spatial hallucination [9] and struggle immensely on multi-tier evaluation frameworks [16] when subjected to multi-hop queries.

1.2. Tool-Augmented Visual Reasoning

To overcome the spatial limitations of frozen LLM weights, frameworks like ViperGPT [8], Visual Programming (VisProg) [7], and ReAct [4] allow LLMs to call discrete APIs to solve tasks compositionally. By generating Chain-of-Thought (CoT) reasoning traces [10], the LLM acts as a logical orchestrator. In geospatial contexts, this means leveraging foundation models like Segment Anything (SAM) [5], whose efficacy on aerial data has been proven by SAMRS [6], and Bitemporal Image Transformers (BIT) [15] for change detection, rather than relying on the LLM’s weak internal spatial representations.

1.3. Visual Attention Masking

A critical failure point in multi-hop VQA is “attention collapse,” where a VLM fixates on a primary anchor object [14]. Inspired by the Hide-and-Seek architecture [13], which forces networks to discover secondary features by physically masking discriminative regions, we introduce Recurrent Visual Attention Masking (RVAM).

2. Methodology and Architectural Evolution

The project was executed in three progressive phases to establish a baseline, identify reasoning bottlenecks, and systematically eliminate those failures.

2.1. Benchmark Generation & Baseline (Phase 1)

To rigorously evaluate spatial reasoning, we developed a highly strict, unbiased 600-question spatial benchmark using the **xView** dataset [12]. Questions were programmatically generated to ensure a perfect 50/50 Yes/No split across 2-Hop, 3-Hop, and 4-Hop queries. Evaluating the **LLaVA-1.5 Baseline** revealed severe hallucination rates. A “Blind Stress Test” yielded positive accuracy, proving the baseline was exploiting linguistic correlations rather than visually grounding its answers.

2.2. Agentic Framework & Perceptual Bottleneck (Phase 2)

To solve these hallucinations, we introduced a **ReAct** loop utilizing LLaVA-1.5, YOLOv8-ORB [11], and SAM. While this grounded the model, it introduced the *Compounding Error Penalty*. If the object detector missed a target due to environmental noise, the agent failed the reasoning task regardless of its logical soundness. Furthermore, our initial RVAM implementation used solid black boxes to mask objects; this caused destructive interference by covering adjacent contextual terrain.

2.3. V2: Bulletproof GeoAgent (Phase 3)

To isolate true reasoning capabilities from predictive detector failures, we developed the final V2 Architecture:

- **Qwen2-VL Transition:** Replaced LLaVA with Qwen2-VL for dynamic resolution scaling and superior instruction-following.
- **Oracle GeoTools:** Replaced YOLO with Oracle tools backed directly by the xView GeoJSON ground truth. This mathematically isolated the VLM’s reasoning logic from perceptual blindness [8].
- **Gaussian RVAM:** Replaced solid black boxes with

a Heavy Gaussian Blur (`cv2.GaussianBlur`). This effectively scrambles discriminative features while preserving surrounding pixels, forcing the ReAct loop to re-evaluate the scene for secondary targets smoothly.

2.4. Final Results and Comparative Analysis

Table 1: Spatial Reasoning Accuracy: Baseline vs. V2 GeoAgent

Architecture	2-Hop	3-Hop	4-Hop
LLaVA-1.5 Baseline	57.0%	59.5%	53.0%
V2 GeoAgent (ReAct+Oracle)	89.5%	83.0%	84.5%

As shown in Table 1, the V2 GeoAgent systematically dismantles the baseline’s limitations. While the baseline suffers from severe step degradation—dropping to nearly random guessing (53.0%) on 4-hop queries—the GeoAgent maintains highly robust logical grounding, achieving **84.5% accuracy on the same 4-hop tasks (a +31.5% absolute improvement)**.

Reasoning Trace Example (V2):

```
Question: Find the Aircraft on the left. Now
move your focus to the right half. Is there
a Cargo Truck located in that right half?

[LLM]: Thought: I need to locate the planes on
the left, but I shouldn't be distracted by
them.
Action: visual_masker Aircraft
[TOOL]: RVAM: Masked 2 'Aircraft' instance(s)
with Gaussian Blur. Attention redirected.
[LLM]: Thought: Now I can scan the right half.
Action: spatial_query Cargo Truck right
[TOOL]: CONFIRMED: 1 Cargo Truck(s) present in
RIGHT half.
[LLM]: Final Answer: Yes
```

3. Conclusion

In this project, we proposed GeoAgent to solve step degradation in remote sensing VLMs. By transitioning to a Qwen2-VL ReAct loop utilizing Oracle-backed tools and Gaussian RVAM, we successfully isolated and mathematically grounded the model’s reasoning capabilities. Achieving an 84.5% accuracy on strict 4-hop spatial tasks, the GeoAgent proves that Agentic AI architectures, verified through rigorous adversarial integrity testing, provide a highly reliable

pathway to verifiable spatial reasoning over complex aerial data.

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